

CEU SAN PABLO
Midterm 214 on 25th March

1. (0.5 points) What is the mean Capture Rate in 2nd Generation?

- a. 98'76
- b. 106'19
- c. 91'90

2. (0.5 points) Which Pokemon has the highest Attack?

Write your answer here.

3. (0.5 points) Draw the scatter/dots of the height as a function of the weight

Remember to send the output after exam.

4. (0.5 points) What is the probability of obtaining exactly 1 Legendary pokemon ?

Write your answer here.

5. (0.5 points) Which generation has the highest mean attack?

- a. First Generation
- b. Second Generation
- c. Third Generation
- d. Seventh Generation

6. (0.5 points) The 50% of Legendary Pokemon have a higher attack than 50% of Non-Legendary Pokemon:

- a. True
- b. False

7. (0.5 points) We want to estimate attack as a function of speed. Is a good of fit?

- a. No, $R^2 = 0'124$ so the 87'6% of the data is not explained
- b. Yes, $R^2 = 0'124$. ¡Very accurated!
- c. No, $R^2 = 0'124$ so the 12'4% of the data is not explained

8. (1 point) What is the Kurtosis of HP? Draw the histogram with the normal curve.

0.5 for kurtosis and 0.5 for histogram. Write your answer here

9. (0.5 points) Which Pokemon have more homogeneous speed: legendary or not?

- a. Non-Legendary
- b. Legendary

10. (0.5 points) What is the skewness of Special Defense?

- a. Right
- b. Left
- c. Symmetric

11. (1.5 points) Draw a histogram of speed comparing legendary and non-legendary Pokemon. Could there be outliers in the legendary Pokemon's speed distribution? If so, identify the greatest one.

Write your answer here.

12. (0.5 points) Which generation has the highest mean speed?

- a. First Generation
- b. Second Generation
- c. 5th Generation
- d. 7th Generation

13. (0.5 points) Compute the quartiles of capture rate

Q1:
Q2:
Q3:

14. (0.5 points) Is the mode of experience growth unique?

- a. Yes
- b. No

15. (0,5 points) What is the equation of linear regression if we want to estimate the attack as a function of speed?

Write your answer here.

16. (0.5 points) Suppose that the base happiness is distributed as a normal. What is the probability of finding a Pokemon with base happiness lower than 40?

- a. 0'902
- b. 0'098
- c. -0'098

17. (0.5 points) What is the best model to predict height in function of weight? What is the equation?

Write your answer here

